

SE-LAB 1

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Problem Statement 34 — Digital Gift Card & Loyalty Point Exchange

1. Requirements Table

Functional Requirements (FR-001 – FR-005)

ID	Priority	Description	Type	Acceptance Criteria	Rationale
FR-001	High	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	Functional	Pass: Voucher code generated with cryptographic checksum. Fail: Expired or duplicate voucher redeemed.	Core conversion pipeline is the foundation of the wallet's value exchange; without validated, checksummed vouchers the platform cannot guarantee redemption integrity.
FR-002	High	The system shall allow an Account Holder to aggregate loyalty points from multiple linked Merchant Partner programs into a single unified wallet balance.	Functional	Pass: Balances from all linked merchant accounts are summed and displayed within one wallet view, updated within 5 minutes of a source update. Fail: Any linked balance is missing, stale beyond 5 minutes, or double-counted.	Aggregation is the platform's primary value proposition; users must trust that a single wallet view accurately reflects points held across brands.
FR-003	Medium	The system shall compute and display the current dynamic exchange rate before an Account Holder confirms a point-to-credit conversion, and shall lock that rate for the duration of the confirmation window.	Functional	Pass: Quoted rate matches the rate applied at confirmation when confirmed within the lock window (60s). Fail: Applied rate differs from the quoted rate, or no rate is shown prior to confirmation.	Because exchange rates are dynamic, users need a guaranteed quote to make an informed decision and to prevent disputes over unexpected conversion outcomes.
FR-004	High	The system shall allow an Account Holder to redeem a generated voucher for a digital gift card, applying an anti-fraud lock that marks the voucher as consumed atomically upon successful redemption.	Functional	Pass: Voucher status transitions to 'redeemed' in the same transaction as gift card issuance; a second redemption attempt is rejected. Fail: Voucher can be redeemed more than once, or gift card is issued without the voucher being locked.	Directly implements the anti-fraud voucher locking called out in the problem statement; prevents double-spend of a single-use financial instrument.
FR-005	Medium	The system shall provide a Merchant Partner dashboard for issuing or revoking loyalty points to Account Holders and for viewing aggregated redemption analytics for their brand.	Functional	Pass: Point issuance/revocation actions reflect in the Account Holder's wallet within 5 minutes and appear in the merchant's analytics view. Fail: Issued/revoked points are not reflected in the wallet, or analytics omit recent redemptions.	Merchant Partners are a primary stakeholder and need self-service tools to manage their loyalty program and measure redemption performance.

Non-Functional Requirements (NFR-001 – NFR-002)

ID	Priority	Description	Type	Acceptance Criteria	Rationale
NFR-001	High	Point conversion and gift card redemption operations must process within 250 ms with zero ledger balance drift.	Performance & Security	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: p95 latency exceeds 250 ms, or ledger reconciliation shows any balance drift.	Users expect near-instant redemption at point of sale; any ledger drift would represent a direct financial integrity failure.
NFR-002	High	The platform shall maintain 99.9% monthly uptime for wallet, conversion, and redemption services, with all point-balance and voucher-state updates committed as atomic transactions.	Reliability & Availability	Pass: Monthly uptime monitoring confirms >= 99.9% availability; chaos/failure-injection tests confirm no partial (non-atomic) balance or voucher updates under simulated service failure. Fail: Uptime falls below 99.9%, or any test reveals a partially-committed transaction.	As a financial exchange platform handling real monetary value (points and gift cards), availability and transactional atomicity are essential to user trust and regulatory expectations.

2. UML Use-Case Diagram

Digital Gift Card & Loyalty Point Exchange System

Actors

- **Account Holder**
- **Merchant Partner**

Use Cases

The **Account Holder** interacts with the following use cases:

- Authenticate Account Holder
- View Loyalty Points
- Aggregate Loyalty Points
- Convert Loyalty Points to Universal Exchange Credits
- View Dynamic Exchange Rate
- View Transaction / Ledger History
- Generate Digital Voucher
- Redeem Digital Gift Card

The **Merchant Partner** interacts with the following use cases:

- Validate Loyalty Points
- Validate Voucher

Included Use Cases

The following «include» relationships are represented in the diagram:

1. **Convert Loyalty Points to Universal Exchange Credits** «include» **View Dynamic Exchange Rate**
2. **Convert Loyalty Points to Universal Exchange Credits** «include» **Validate Loyalty Points**
3. **Redeem Digital Gift Card** «include» **Validate Voucher**

4. **Generate Digital Voucher** «include» **Apply Anti-Fraud Voucher Lock**

Extended Use Case

The diagram contains the following «extend» relationship:

- **Apply Anti-Fraud Voucher Lock** «extend» **Redeem Digital Gift Card**

The anti-fraud voucher locking behavior is triggered as an additional condition during the digital gift-card redemption process.

Use-Case Diagram Description

The UML Use-Case Diagram represents the interaction between the **Account Holder**, **Merchant Partner**, and the **Digital Gift Card & Loyalty Point Exchange System**.

The Account Holder can authenticate their account, view and aggregate loyalty points, convert loyalty points into universal exchange credits using the dynamic exchange rate, view transaction or ledger history, generate digital vouchers, and redeem digital gift cards.

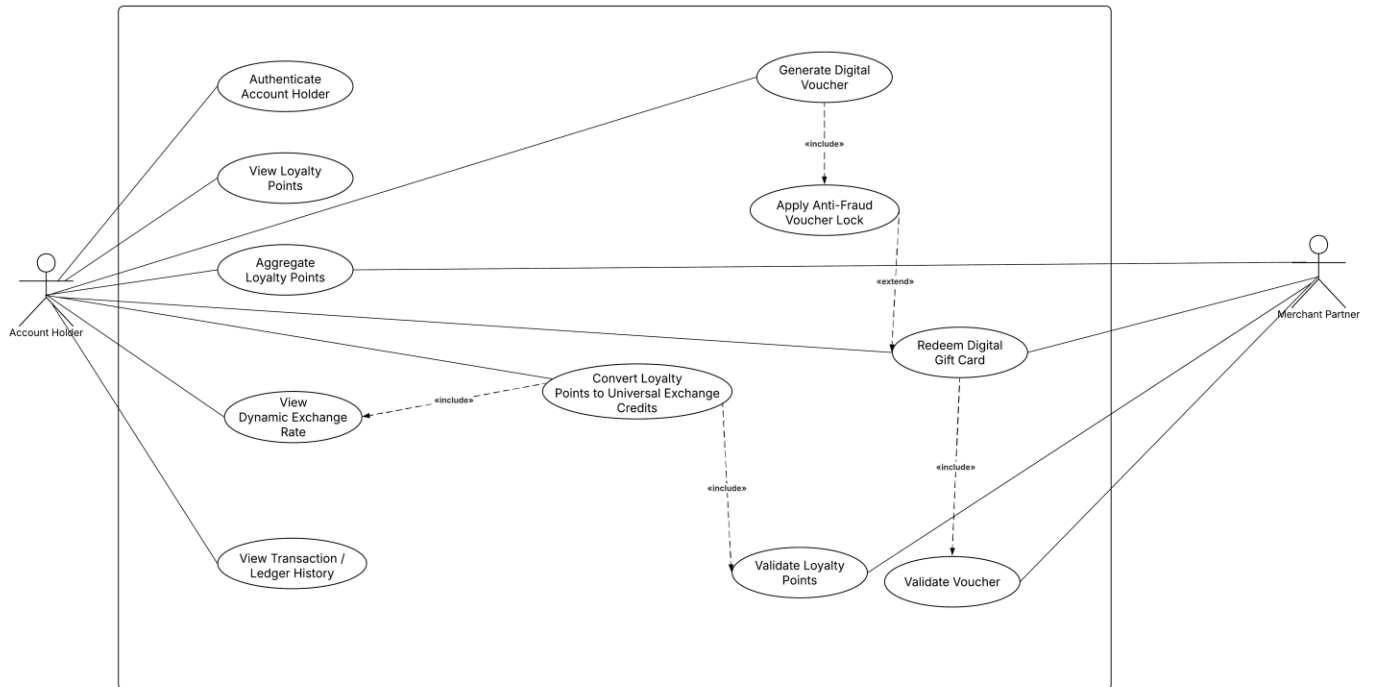
The Merchant Partner participates in the validation process by validating loyalty points and digital vouchers.

The «include» relationships represent mandatory supporting behavior. Conversion of loyalty points includes viewing the dynamic exchange rate and validating loyalty points. Redemption of a digital gift card includes voucher validation. Generation of a digital voucher includes applying the anti-fraud voucher lock.

The «extend» relationship represents additional anti-fraud behavior associated with redemption. **Apply Anti-Fraud Voucher Lock** extends **Redeem Digital Gift Card**, providing an additional security mechanism during the redemption process.

The diagram models the two required actors, the major system use cases, and both «include» and «extend» relationships required for the assignment.

UML Use-Case Diagram – Digital Gift Card & Loyalty Point Exchange



3. Use-Case Flow Specification

Use Case: Redeem Digital Gift Card

Primary Actor: Account Holder

Supporting Actor: Merchant Partner

Preconditions

- The Account Holder is authenticated and logged into the digital wallet platform.
- The Account Holder has a valid, non-expired, non-redeemed voucher.
- The Account Holder has sufficient universal exchange credits to redeem the selected digital gift card.
- The voucher is associated with a valid loyalty-point conversion.

Postconditions

Success:

- The voucher is marked as **redeemed** and locked against reuse.
- A digital gift card is issued to the Account Holder.
- The exchange-credit ledger balance is decremented correctly.
- The redemption transaction is recorded in the system.

Failure:

- No digital gift card is issued.

- The voucher remains active or is placed in a flagged state if suspicious activity is detected.
- The Account Holder's exchange-credit balance remains unchanged.

Main Success Scenario

1. The Account Holder selects a valid voucher and chooses **Redeem Digital Gift Card**.
2. The system authenticates the Account Holder.
3. The system validates the voucher code, including its validity, expiry status, and redemption status.
4. The system confirms that the voucher is valid, unexpired, and has not been previously redeemed.
5. The system checks that sufficient exchange credits are available for the selected gift card.
6. The system securely locks the voucher against reuse and processes the redemption.
7. The system deducts the required exchange credits from the Account Holder's ledger.
8. The system generates and displays the digital gift card and redemption confirmation.
9. The system records the redemption transaction for future tracking and Merchant Partner reference.

Alternate Flow A1: Suspicious Redemption Detected

1. During voucher validation or redemption, the system detects suspicious or anomalous transaction activity.
2. The system triggers **Apply Anti-Fraud Voucher Lock** as an extended security behavior.
3. The voucher is placed into a flagged/hold state instead of being redeemed.
4. The system denies the redemption and informs the Account Holder that additional verification is required.
5. No gift card is issued and no exchange credits are deducted.
6. The transaction is recorded for fraud review.